

## Impact-Aware State Estimation for Off-Road Motion

GPS / IMU / Camera Fusion with Adaptive NHC Relaxation via Jerk Heuristic and LSTM on SE<sub>2</sub>(3)

Pranav Jain · Avantika Rattan | Department of Robotics, University of Michigan | {psjain, arattan}@umich.edu

KEY RESULT

**-96%**

ATE Reduction

6.33 m → 0.258 m mean across 6 sequences

### 01 · MOTIVATION & PROBLEM

Off-road vehicles face dynamics that violate two core assumptions of standard GPS/IMU filters:

- Stationary noise: Rocks, wheel bounce & hills generate 500–3000 m/s<sup>3</sup> jerk (60–375× gravity). Fixed Q is violated continuously.
- Non-Holonomic Constraint: Asserting v<sub>lat</sub> = v<sub>vert</sub> = 0 is physically violated on any slope or impact event.
- Critical failure: Hard NHC actively CONFLICTS with GPS corrections on hilly terrain → catastrophic filter oscillation.

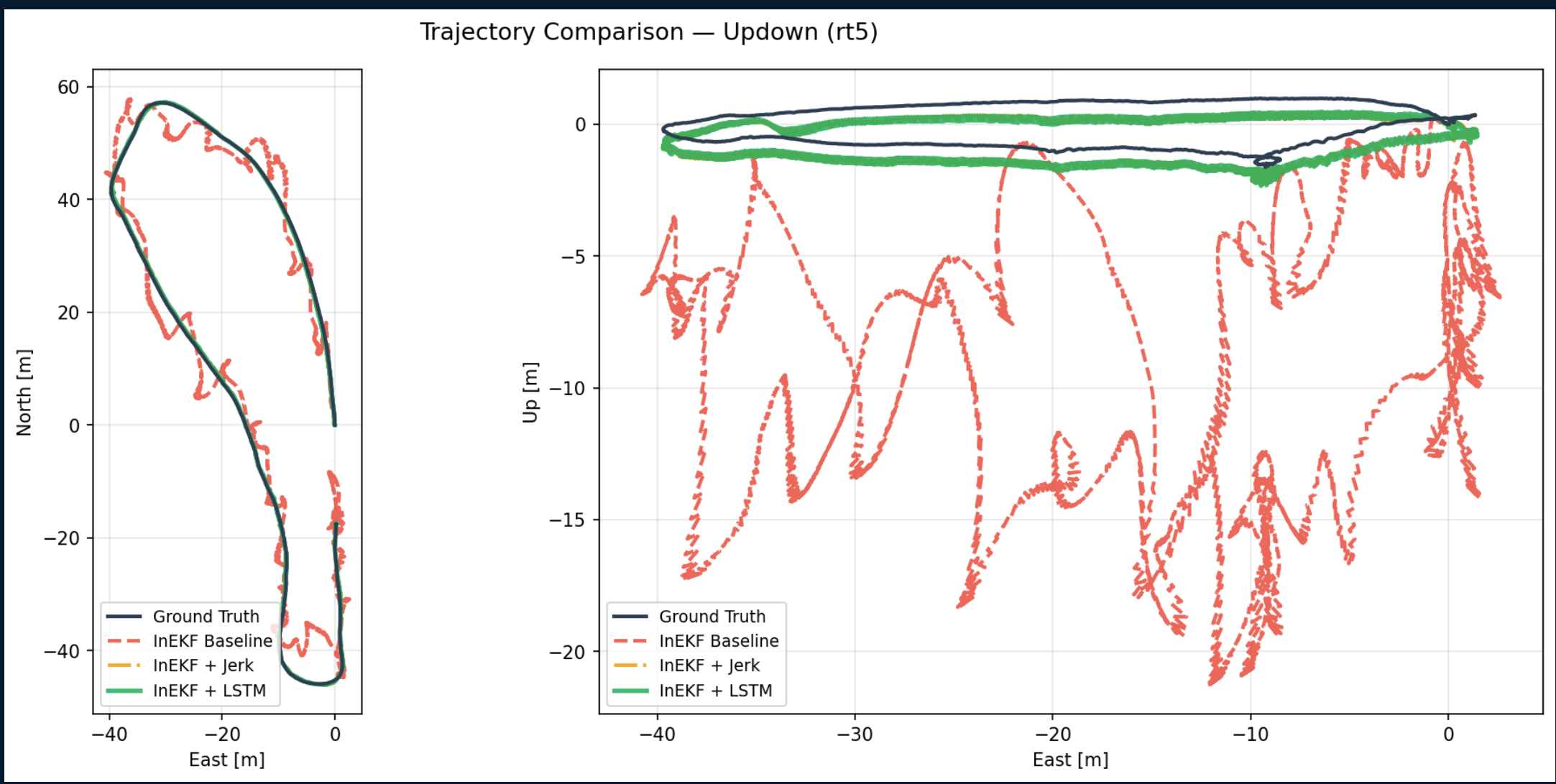
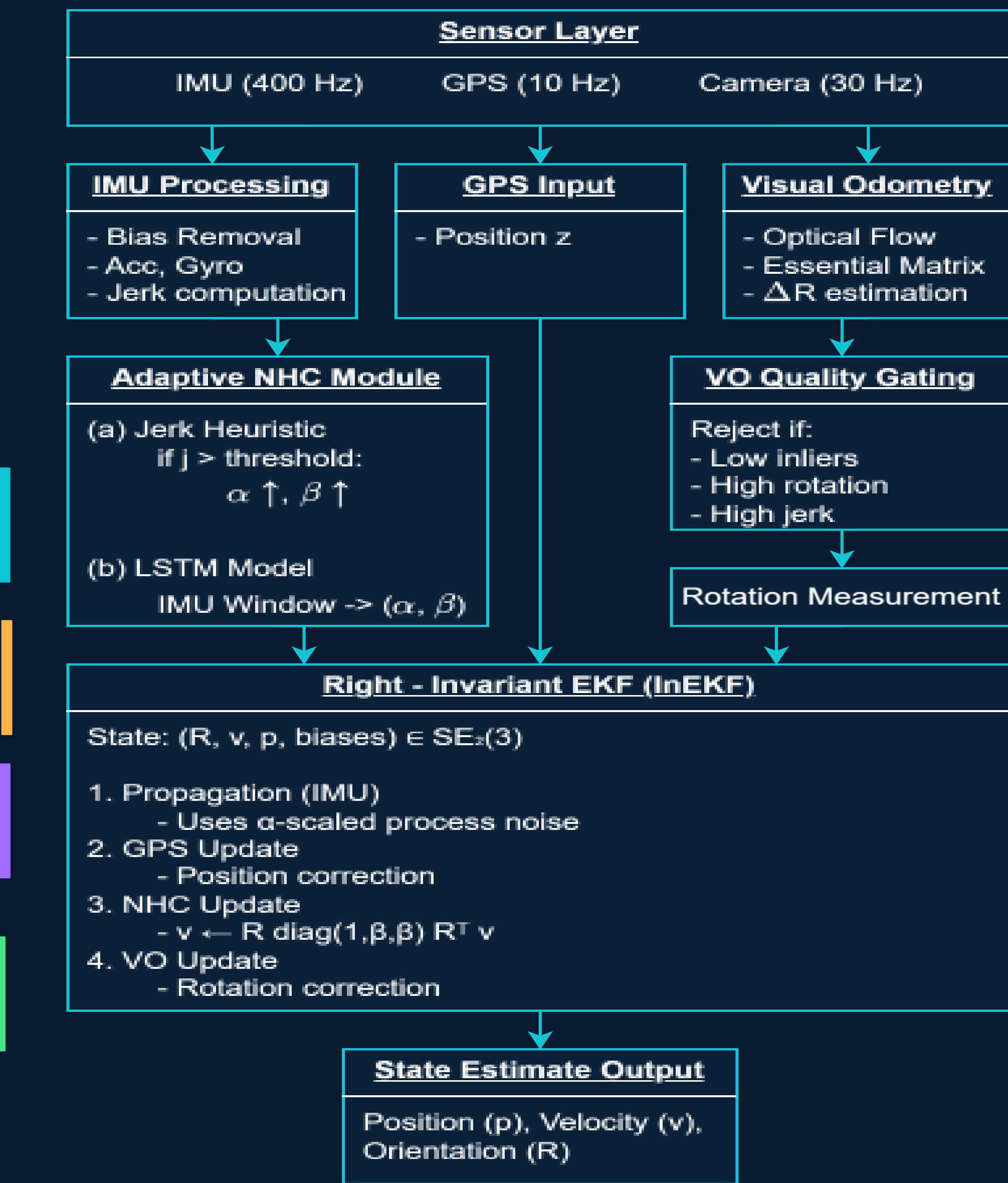
#### The NHC–GPS Conflict



Fix:  $\beta \in [0,1]$  relaxation factor allows GPS corrections to persist. Hard NHC ( $\beta=0$ ) is replaced by adaptive soft constraint.

📍 ROOAD Dataset — Clearpath Warthog UGV | VN300 IMU @ 400 Hz | RTK GPS @ 10 Hz (2 cm) | Basler Camera @ 30 Hz

### 02 · SYSTEM ARCHITECTURE — INEKF ON SE<sub>2</sub>(3)



### 03 · ADAPTIVE MECHANISMS

| InEKF + Jerk                                                                  | InEKF + LSTM                                    |
|-------------------------------------------------------------------------------|-------------------------------------------------|
| Smoothed 20-sample jerk $j(t)$                                                | 50-sample window $\times$ 9 features            |
| $j > 8 \text{ m/s}^3 \rightarrow \alpha \leftarrow 15, \beta \leftarrow 0.85$ | 2-layer LSTM, hidden=128, drop=0.2              |
| Decay: $\alpha=0.95, \beta=0.90$ per step                                     | Head $\alpha$ : sigmoid $\rightarrow [0.1, 20]$ |
| No training required                                                          | Head $\beta$ : sigmoid $\rightarrow [0, 0.85]$  |
| ⚠ Saturates at $\alpha=15$ throughout                                         | ✓ Modulates $\alpha \in [1,20]$ continuously    |

#### LSTM Architecture

| Field         | Detail                                                                            |
|---------------|-----------------------------------------------------------------------------------|
| Input         | (batch, 50, 9) — [ax, ay, az, gx, gy, gz, jerk,  a ,  w ]                         |
| LSTM          | 2 layers · hidden=128 · dropout=0.2 · CUDA                                        |
| Head $\alpha$ | Linear $\rightarrow$ Sigmoid $\rightarrow \alpha \in [0.1, 20]$ (Q scale)         |
| Head $\beta$  | Linear $\rightarrow$ Sigmoid $\rightarrow \beta \in [0, 0.85]$ (NHC relax)        |
| Loss          | $\text{MSE}(\log \hat{\alpha}, \log \alpha^*) + \text{MSE}(\hat{\beta}, \beta^*)$ |
| Train         | 4 seqs · 7,323 windows · 200 epochs · val=0.058                                   |
| Test          | rt5_rim, rt5_updown (never seen during training)                                  |

#### VO Pipeline

|                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FAST corners $\rightarrow$ Lucas-Kanade optical flow (bidirectional check) $\rightarrow$ RANSAC Essential matrix $\rightarrow$ recoverPose $\rightarrow$ $\Delta R_{\text{cam}} \rightarrow$ body frame via $R_{\text{cb}} \rightarrow$ EKF rotation update |
| Quality gates: inliers > 150   rotation < 15°   jerk < 500 m/s <sup>3</sup>                                                                                                                                                                                 |

### 04 · QUANTITATIVE RESULTS — ATE & RPE [M]

| Sequence     | VO | Baseline ATE | Jerk ATE | LSTM ATE | RPE Jerk | RPE LSTM |
|--------------|----|--------------|----------|----------|----------|----------|
| rt4_gravel   | —  | 10.97        | 0.266    | 0.288    | 0.100    | 0.098    |
| rt4_rim      | —  | 3.295        | 0.231    | 0.176    | 0.091    | 0.083    |
| rt4_updown   | Y  | 8.014        | 0.297    | 0.299    | 0.100    | 0.100    |
| rt5_gravel   | —  | 6.471        | 0.292    | 0.318    | 0.100    | 0.095    |
| rt5_rim †    | —  | 3.725        | 0.203    | 0.312    | 0.085    | 0.086    |
| rt5_updown † | Y  | 5.482        | 0.261    | 0.270    | 0.099    | 0.097    |
| Mean         | —  | 6.326        | 0.258    | 0.277    | 0.096    | 0.093    |

† Test sequences — unseen during training | ⬆ Bold green = best per row | Y = Visual Odometry active

### 05 · PERFORMANCE SUMMARY

|                                                                    |                                                                                |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>96%</b><br>ATE Reduction<br>6.33 m $\rightarrow$ 0.26 m         | <b>0.26 m</b><br>Best ATE<br>InEKF+Jerk rt5_rim                                |
| <b>95%</b><br>Speed Improvement<br>2.4 $\rightarrow$ 0.10 m/s RMSE | <b>55° <math>\rightarrow</math> 3°</b><br>Heading Error<br>rt5_updown (GPS+VO) |

### 06 · VISUAL ODOMETRY EXTENSION

Monocular VO adds rotation constraints at 30 Hz between GPS fixes using LK optical flow + RANSAC Essential matrix decomposition.

| Sequence   | Filter     | GPS only | GPS+VO | $\Delta$ ATE |
|------------|------------|----------|--------|--------------|
| rt4_updown | InEKF+Jerk | 0.255    | 0.297  | +16% ⬆       |
| rt4_updown | InEKF+LSTM | 0.263    | 0.299  | +14% ⬆       |
| rt5_updown | InEKF+Jerk | 0.298    | 0.261  | -12% ⬆       |
| rt5_updown | InEKF+LSTM | 0.334    | 0.270  | -19% ⬆       |

Mixed results due to approximate  $R_{\text{cb}}$  extrinsic. Exact Kalibr calibration would resolve inconsistency. VO tracking: 68–73% success rate.

### 07 · FUTURE WORK

|                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online Extrinsic Calibration</b><br><br>Augment InEKF state with $\delta \phi_{\text{cb}} \in \mathbb{R}^3$ (18D). VO Jacobian gains chain-rule block $\partial \text{innov} / \partial \delta \phi_{\text{cb}}$ . GPS+VO jointly self-calibrate $R_{\text{cb}}$ without separate calibration sequence. |
| <b>Terrain-Conditioned NHC</b><br><br>Add MiDaS depth slope $\theta$ as LSTM context feature. On slopes $\beta \sim 0.85$ even without large jerk. Decouples terrain geometry from impact magnitude.                                                                                                       |
| <b>Observability Analysis</b><br><br>Identify conditions for extrinsic identifiability. Bounded prior $\ \delta \phi_{\text{cb}}\  < 20^\circ$ prevents divergence on straight-line motion. Validate against Kalibr GT.                                                                                    |

### ◆ CORE FINDING

|                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| The hard NHC is not merely suboptimal — it actively BREAKS GPS/IMU fusion by conflicting with position corrections on non-flat terrain. |
| Adaptive relaxation ( $\beta > 0$ ) is a prerequisite, not a refinement, for robust off-road GPS/IMU state estimation.                  |

